Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.8

1. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, and define $T : \mathbb{R}^2 \to \mathbb{R}^2$ by $T(\mathbf{x}) = A\mathbf{x}$. Find the images under T of $\mathbf{u} = \begin{bmatrix} 4 \\ -6 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$.

$T(\mathbf{u}) = \begin{bmatrix} 8 \\ -12 \end{bmatrix}$

$T(\mathbf{v}) = \begin{bmatrix} 2a \\ 2b \end{bmatrix}$

2. If T is defined by $T(\mathbf{x}) = A\mathbf{x}$, find a vector $\mathbf{x}$ whose image under T is $\mathbf{b}$, and determine whether $\mathbf{x}$ is unique. Let $A = \begin{bmatrix} 1 & -3 & 5 \\ 0 & 1 & -5 \\ 3 & -10 & 15 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -6 \\ -8 \\ -5 \end{bmatrix}$.

Find a single vector $\mathbf{x}$ whose image under T is $\mathbf{b}$.

$\mathbf{x} = \begin{bmatrix} -40 \\ -13 \\ -1 \end{bmatrix}$

Is the vector $\mathbf{x}$ found in the previous step unique?

- **A.** Yes, because there is a free variable in the system of equations.
- **B.** Yes, because there are no free variables in the system of equations.
- **C.** No, because there are no free variables in the system of equations.
- **D.** No, because there is a free variable in the system of equations.

3. If T is defined by $T(\mathbf{x}) = A\mathbf{x}$, find a vector $\mathbf{x}$ whose image under T is $\mathbf{b}$, and determine whether $\mathbf{x}$ is unique. Let $A = \begin{bmatrix} 1 & -7 & -19 \\ -4 & 19 & 49 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$.

Find a single vector $\mathbf{x}$ whose image under T is $\mathbf{b}$.

$\mathbf{x} = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$

Is the vector $\mathbf{x}$ found in the previous step unique?

- **A.** No, because there are no free variables in the system of equations.
- **B.** Yes, because there is a free variable in the system of equations.
- **C.** Yes, because there are no free variables in the system of equations.
- **D.** No, because there is a free variable in the system of equations.

4. Let A be a 6×4 matrix. What must a and b be in order to define $T : \mathbb{R}^a \to \mathbb{R}^b$ by $T(\mathbf{x}) = A\mathbf{x}$?

a = 4
(Simplify your answer.)

b = 6
(Simplify your answer.)

5. How many rows and columns must a matrix A have in order to define a mapping from $\mathbb{R}^4$ into $\mathbb{R}^7$ by the rule $T(\mathbf{x}) = A\mathbf{x}$?

Choose the correct answer below.

- **A.** The matrix A must have 4 rows and 4 columns.
- **B.** The matrix A must have 7 rows and 7 columns.
- **C.** The matrix A must have 4 rows and 7 columns.
- ✔ **D.** The matrix A must have 7 rows and 4 columns.

6. Let $\mathbf{b} = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$, and let A be the matrix $\begin{bmatrix} 1 & -4 & 5 & -5 \\ 0 & 1 & -2 & 5 \\ 3 & -10 & 11 & -4 \end{bmatrix}$. Is **b** in the range of the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$? Why or why not?

Is **b** in the range of the linear transformation? Why or why not?

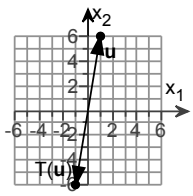
- ✔ **A.** Yes, **b** is in the range of the linear transformation because the system represented by the augmented matrix [A **b**] is consistent.
- **B.** No, **b** is not in the range of the linear transformation because the system represented by the augmented matrix [A **b**] is inconsistent.
- **C.** Yes, **b** is in the range of the linear transformation because the system represented by the augmented matrix [A **b**] is inconsistent.
- **D.** No, **b** is not in the range of the linear transformation because the system represented by the augmented matrix [A **b**] is consistent.

7. Use a rectangular coordinate system to plot $\mathbf{u} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, and their images under the given transformation T. Describe geometrically what T does to each vector $\mathbf{x}$ in $\mathbb{R}^2$.

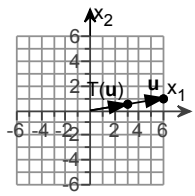
$$T(\mathbf{x}) = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Which graph below shows **u** and its image under the given transformation?

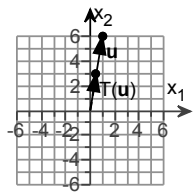
A.



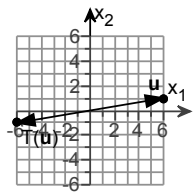
B.



C.

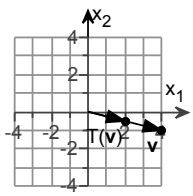


D.

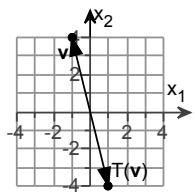


Which graph below shows **v** and its image under the given transformation?

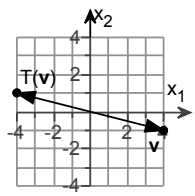
A.



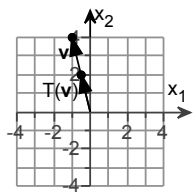
B.



C.



D.



What does T do geometrically to each vector $\mathbf{x}$ in $\mathbb{R}^2$?

- **A.** A projection onto the x_1-axis
- **B.** A reflection through the origin
- **C.** A contraction by the factor of 0.5
- **D.** A rotation about the origin

8. Let T: $\mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation that maps $\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ into $\begin{bmatrix} 6 \\ 1 \end{bmatrix}$ and maps $\mathbf{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$ into $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$. Use the fact that T is linear to find the images under T of $2\mathbf{u}$, $4\mathbf{v}$, and $2\mathbf{u} + 4\mathbf{v}$.

The image of $2\mathbf{u}$ is $\begin{bmatrix} 12 \\ 2 \end{bmatrix}$.

The image of $4\mathbf{v}$ **is** $\begin{bmatrix} -4 \\ 12 \end{bmatrix}$.

The image of $2\mathbf{u} + 4\mathbf{v}$ is $\begin{bmatrix} 8 \\ 14 \end{bmatrix}$.

9. Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $\mathbf{v}_1 = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$, and $\mathbf{v}_2 = \begin{bmatrix} -9 \\ 8 \end{bmatrix}$, and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation that maps $\mathbf{x}$ into $x_1\mathbf{v}_1 + x_2\mathbf{v}_2$. Find a matrix A such that $T(\mathbf{x})$ is $A\mathbf{x}$ for each $\mathbf{x}$.

$A = \begin{bmatrix} 6 & -9 \\ 4 & 8 \end{bmatrix}$

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10. Determine whether the statement below is true or false. Justify the answer.

A linear transformation is a special type of function.

Choose the correct answer below.

- **A.** The statement is true. A linear transformation is a function from $\mathbb{R}$ to $\mathbb{R}$ that assigns to each vector $\mathbf{x}$ in $\mathbb{R}$ a vector $T(\mathbf{x})$ in $\mathbb{R}$.
- **B.** The statement is false. A linear transformation is not a function because it maps one vector $\mathbf{x}$ to more than one vector $T(\mathbf{x})$.
- ✔ **C.** The statement is true. A linear transformation is a function from $\mathbb{R}^n$ to $\mathbb{R}^m$ that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.
- **D.** The statement is false. A linear transformation is not a function because it maps more than one vector $\mathbf{x}$ to the same vector $T(\mathbf{x})$.

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11. Determine whether the statement below is true or false. Justify the answer.

If A is an $m \times n$ matrix, then the range of the transformation $\mathbf{x} \mapsto A\mathbf{x}$ is $\mathbb{R}^m$.

Choose the correct answer below.

- ✔ **A.** The statement is false. The range of the transformation is the set of all linear combinations of the columns of A, because each image of the transformation is of the form $A\mathbf{x}$.
- **B.** The statement is true. The range of the transformation is $\mathbb{R}^m$, because each vector in $\mathbb{R}^m$ is a linear combination of the rows of A.
- **C.** The statement is true. The range of the transformation is $\mathbb{R}^m$, because each vector in $\mathbb{R}^m$ is a linear combination of the columns of A.
- **D.** The statement is false. The range of the transformation is $\mathbb{R}^n$ because the domain of the transformation is $\mathbb{R}^m$.